INTELLIGENT INFORMATION SYSTEMS

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Climate-Related Text Classification for Annual Reports

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2023-03-14

Content

1. Introduction to the data
2. Four approaches to build NLP models
3. Evaluations
4. Data boosting
5. Conclusion

From the first team meeting to final results

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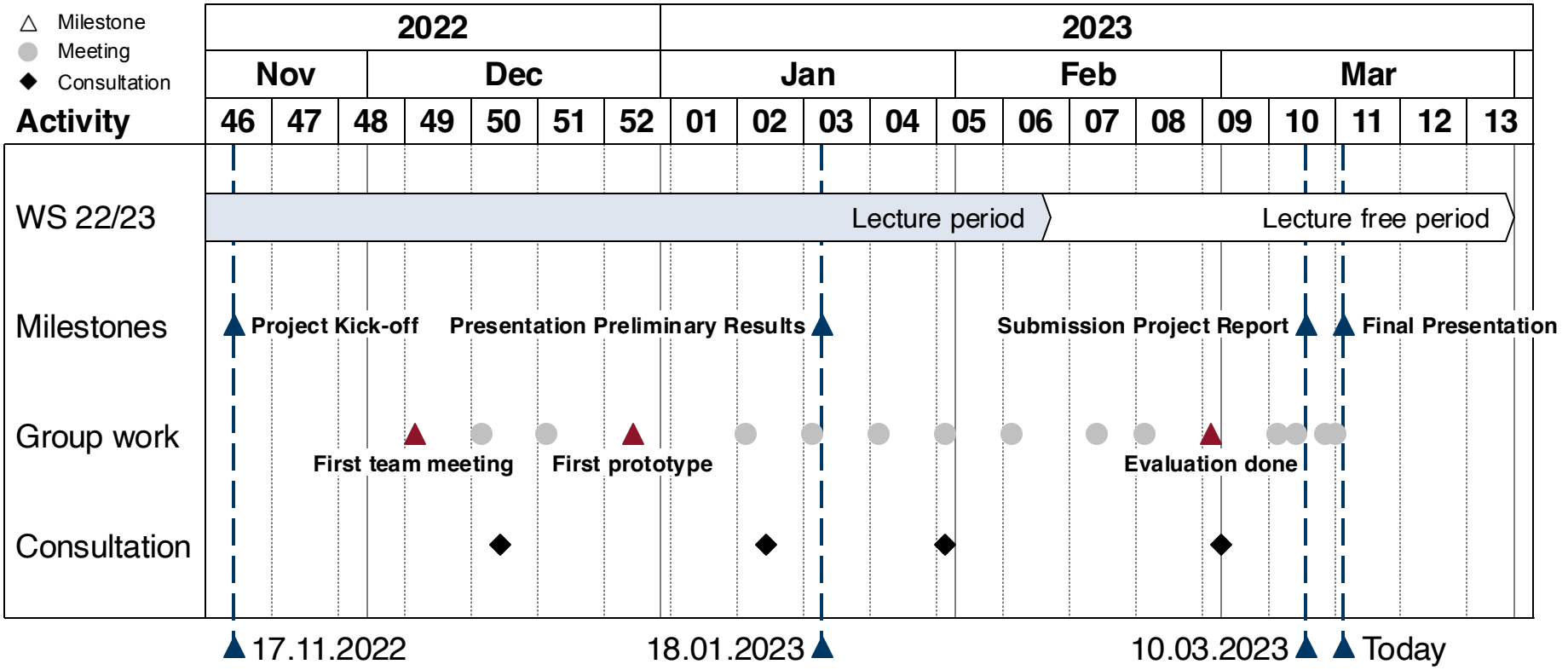
Project Timeline



Data Understanding

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Two Characteristics of Train-set and Validation-set

- 400 labeled paragraphs (initial training data) & 400 labeled paragraphs (validation data)
- 1 million unlabeled paragraphs

1. Imbalanced Data

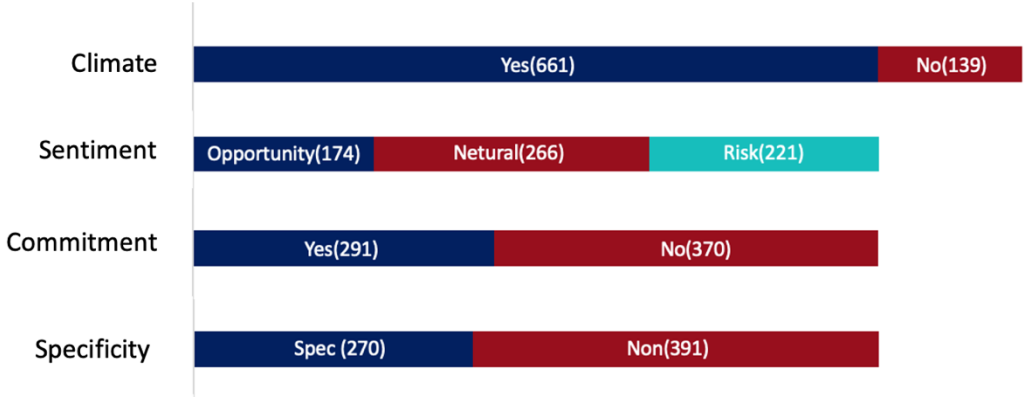
- Imbalanced data can lead to a bias towards the majority class and poor performance on the minority class → Macro average F1 score is adopted

2. Three labels only exist when the first label is marked as "Yes"

- "Climate" is the only label having 800 rows of paragraphs in two datasets.

Four Models :

- Rule-based model
- Bag-of-words model
- Word embedding model
- Transfer learning



Rule-Based NLP System

Steps taken in building Rule-Based system

Rule-Based NLP System

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01 Build Baseline Model

- Approach 1:
  - Build Dictionaries
  - Build Semantic Rules
- Approach 2:
  - Consider most frequent Words

02 Build Dictionaries & Rules

- Approach 1:
  1. WordNet
  2. Sentence Patterns

03 Pre-processing

1. Lowercasing
2. Stop word removal

3.1 Stemming

3.2 Lemmatization

04 Finding Threshold for Models

- Best Pre-processing
- Finding Threshold

05 Model Type (Cross-validated)

- Compare Model's Performance
- Best Model – Approach 2

Build baseline model | Build Dictionaries and Rules

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Rule-Based NLP System

01 02

Approach 1:

1. **Build dictionaries for 4 classifications**
   1. Collect words from
      1. Train-set
      2. Newspapers / Articles
      3. WordNet
      → Find the synonyms & antonyms

➔ **Expand the dictionaries base**

Climate	Sentiment
Natural / Environment words: (e.g. energy/ air/ sea/ soil/ rain/ fuel…) **Impacts on Climate-change:** (e.g. wildfire / flood / emission / greenhouse…)	Opportunity / Risk: **Postive / negative adjectives on environment / business** (+) clean / flexible / innovative… (-) uncertain / vulnerable / badly... **Impact on environment / business** (+) improve / strengthening / efficient… (-) pollution / typhoon / fluctuations…
Commitment	**Specificity**
Commitment synonyms: (e.g. pledged / committed / launched / seeking / facilitate…) **Goal synonyms:** (e.g. support / objective / launched / scheduled / action..)	**Specific events / developments:** (e.g. FY2020, EBRD, ARENA….) **Action to achieve objective:** (e.g. invest / involved / activities / welfare….)

Build Dictionaries and Rules

Rule-Based NLP System

01 02

Approach 1:

2. Semantic Rules for "Specificity" & "Commitment"

1. Specificity:
   1. Contains currency symbol or percent / „%"
   2. Self-defined Specificity phrases
   3. Contains number
2. Commitment:
   1. Contains dollar symbol or ['next', 'by', 'in', 'between', 'period'] before digits
   2. Contains year

- The Fund updated the guidelines for its **$**1.5 billion Sustainable Investment Program (SIP)……

- **In** the **2019** financial year, the Group announced two portfolio transition commitments: - ……aligned with Paris Agreement goals of 45% reduction in emissions **by 2030** and net zero emissions **by 2050.**

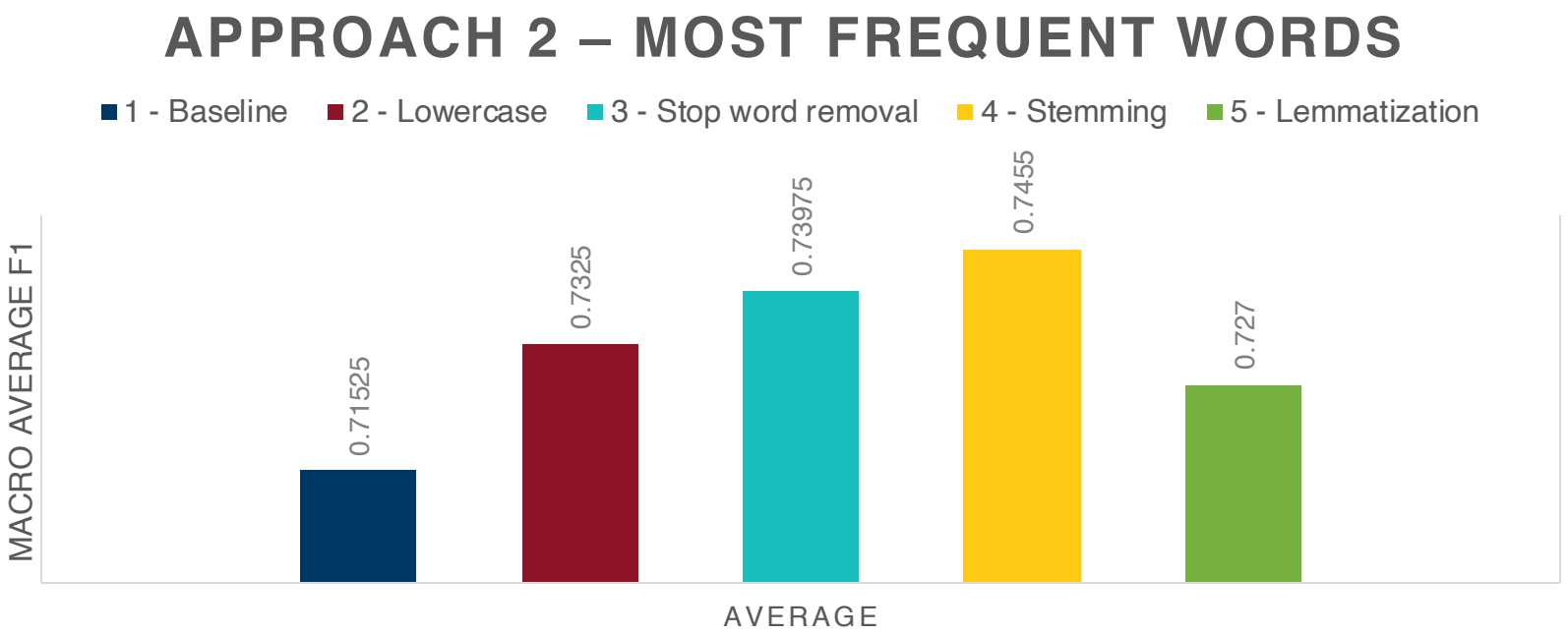
→ difficulties in defining **ambiguous words** and **complex patterns**

→ Approach 2 was adopted

Pre-processing

Rule-Based NLP System

02 03



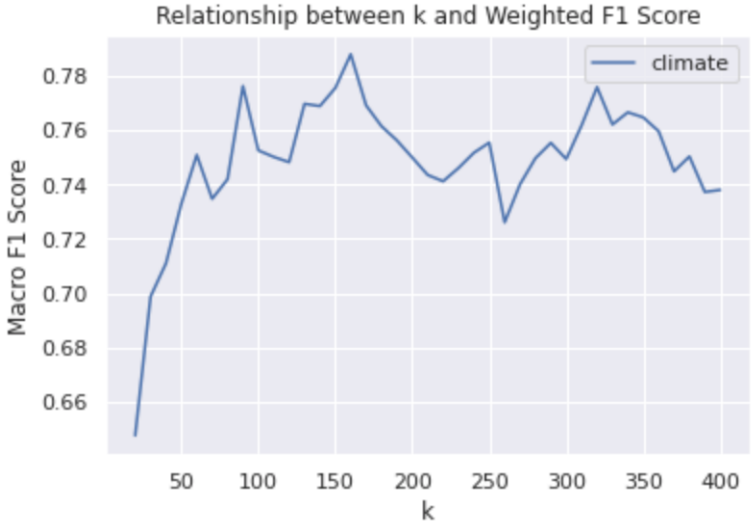
Setting Threshold for Models

Rule-Based NLP System

03 04

Approach 2:

- How many words from the most frequent ones should we use?
  - Decision for optimal number was based on the Marco Average F1 Score

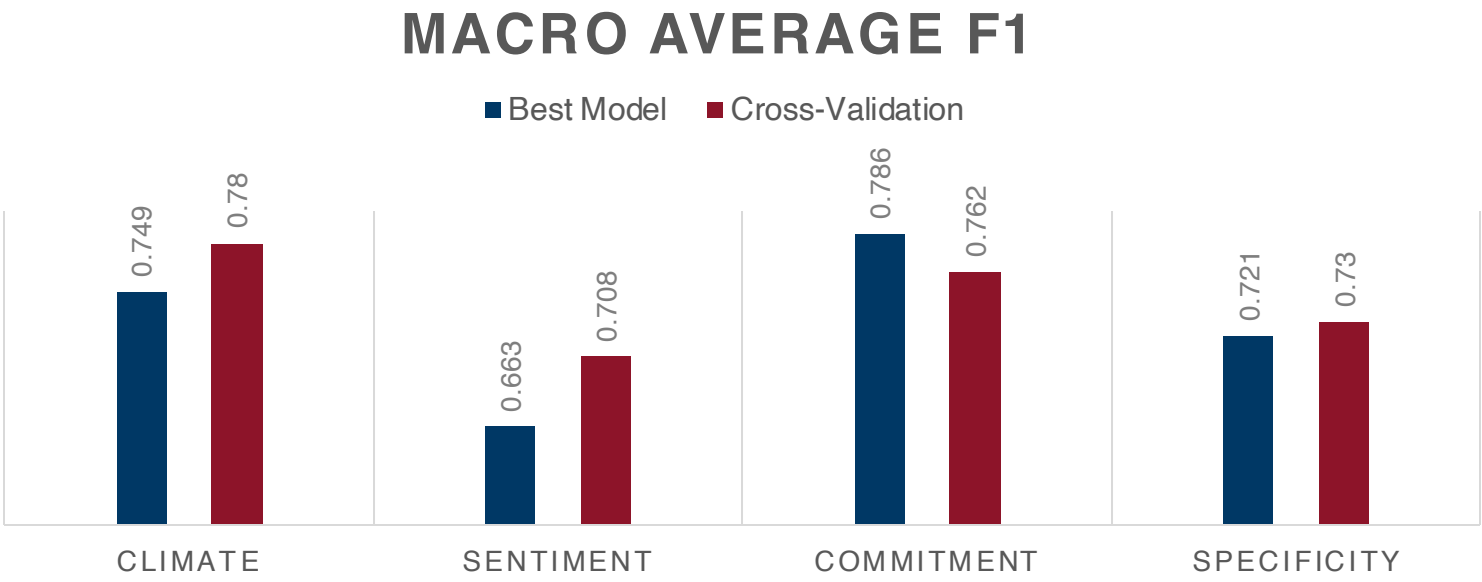


Best Model | Model type (Cross-validated)

Rule-Based NLP System

04 05

Best Model: Approach 2 (Considering most frequent words)



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Bag-of-words based NLP system

Steps taken in building Bag-of-words NLP system

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Bag-of-words based NLP system

01 Pre-processing

1. Stemming
2. Lemmatization
3. Alphabetic Symbols
4. Stop Words Removal

02 Model Type

1. Bag-of-Words
2. TF-IDF
3. N-gram (ngram_range = (1, 2))
4. N-gram (ngram_range = (2, 2))

03 Classifiers

1. Logistic Regression
2. LASSO
3. Ridge
4. SVM

04 Cross-Validation

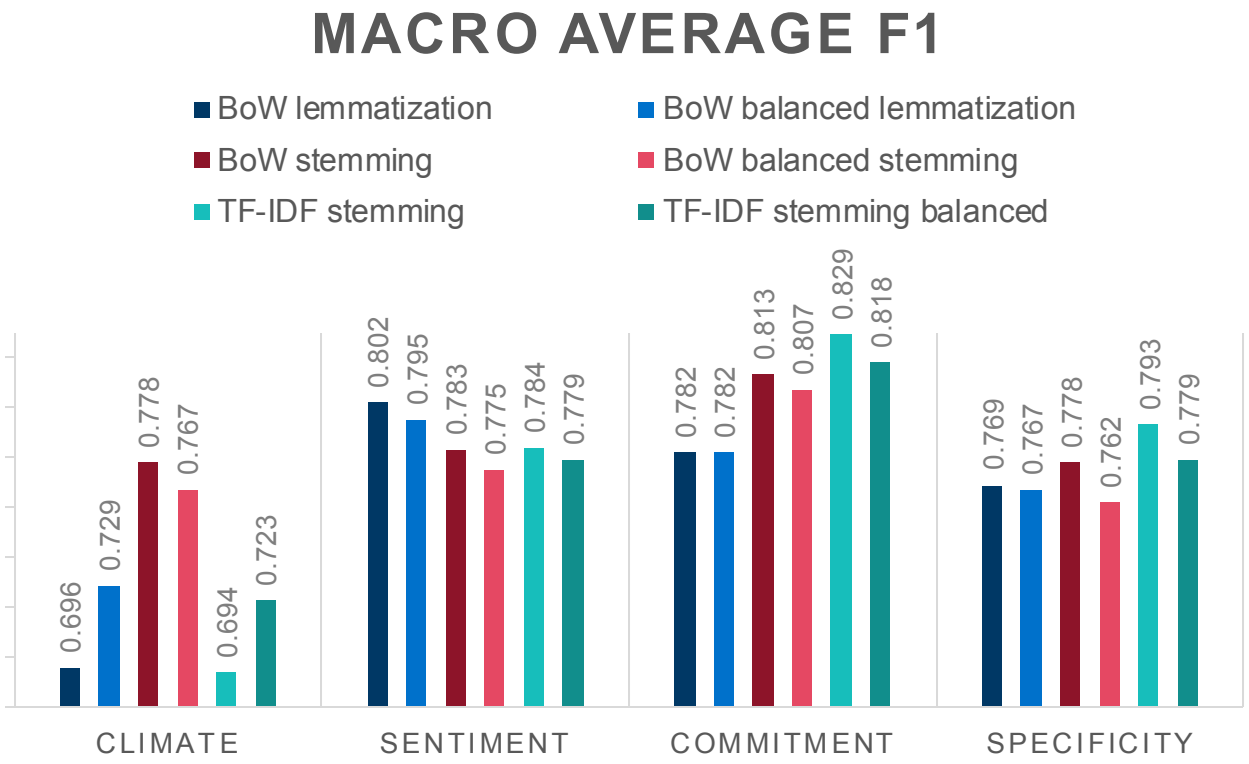
- Best model from previous steps

Bag-of-words Model

Comparison of BoW and TF-IDF

01 02 03

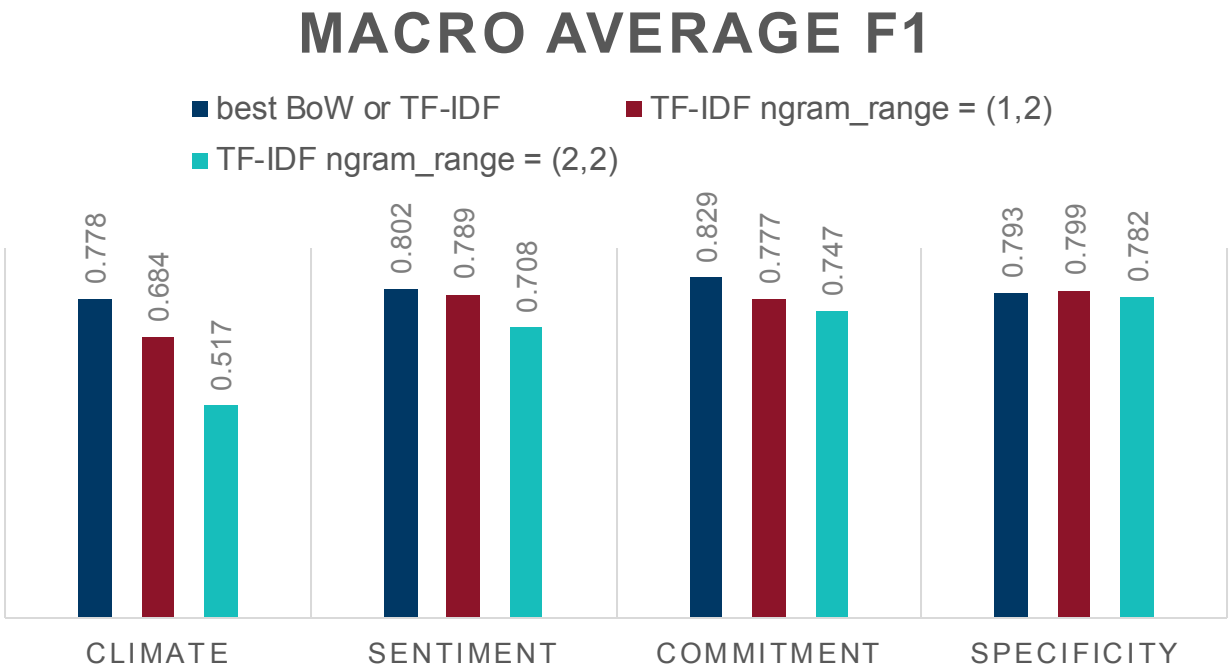
- Stemming > Lemmatization (except climate)
- Non-balanced > Balanced (except climate)
- TF-IDF > BoW (except climate)



Bag-of-words Model

Comparison of best BoW or TF-IDF with N-gram

01 02 03

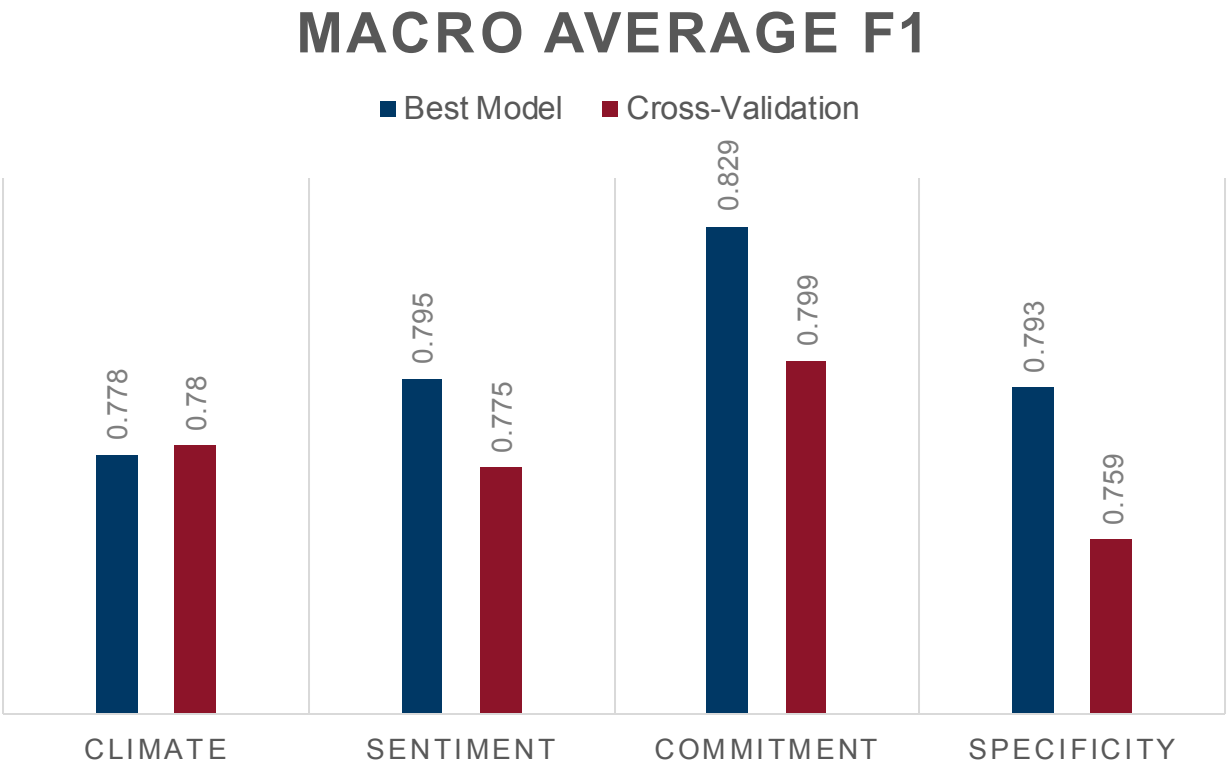


- N-gram is generally worse than BoW and TF-IDF

Bag-of-words Model

Best Result and Cross-validation score

- Climate: BoW stemming with logistic regression
- Sentiment: balanced BoW lemmatization with ridge
- Commitment: TF-IDF stemming with SVM
- Specificity: TF-IDF stemming with SVM



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Word embedding based NLP system

Steps taken in building word embedding based NLP system

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Word embedding based NLP system

01 Build Baseline Model

- Glove Word Embedding
- Transformer TransfoXLTokenizer
- Logistic Regression

02 Pre-processing (Sequential)

1. Link Removal
2. Alphabetic Symbols, Lowercasing
3. Alphanumeric Symbols
4. Special Characters
5. Stop Word Removal

6.1 Stemming
6.2 Lemmatization

03 Pre-processing (Individual)

1. Alphabetic Symbols, Lowercasing
2. Alphanumeric Symbols
3. …

04 Tokenization

- Best Pre-processing
- Tokenization
  - Whitespace Tokenization
  - NLTK Tokenization
  - Transformer Tokenization

05 Word Embedding

- Best from previous steps
- Word Embedding
  - GloVe
  - Self-trained

06 Model type (Cross-validated)

- Best from previous steps
- Model
  - Unoptimized LR
  - Optimized LR
  - Neural Network

Pre-processing (Individual)

Word embedding based NLP system

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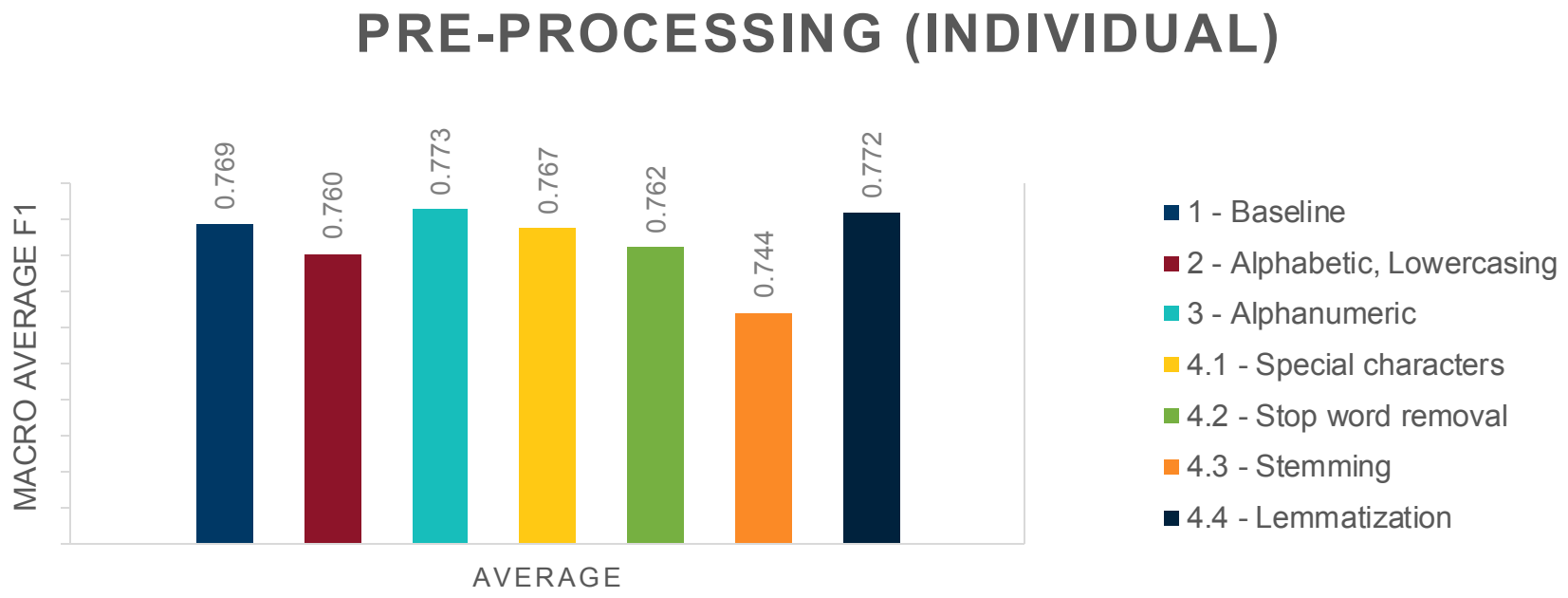

Pre-processing (Individual)

Word embedding based NLP system

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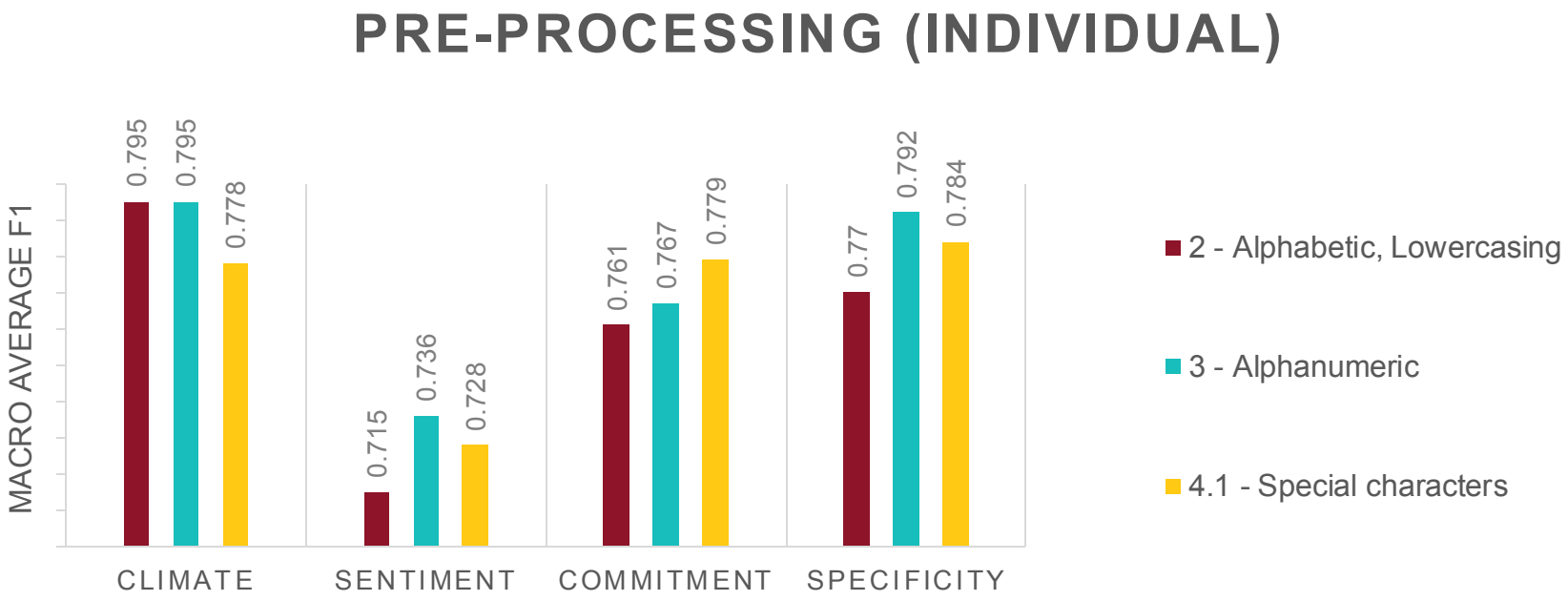
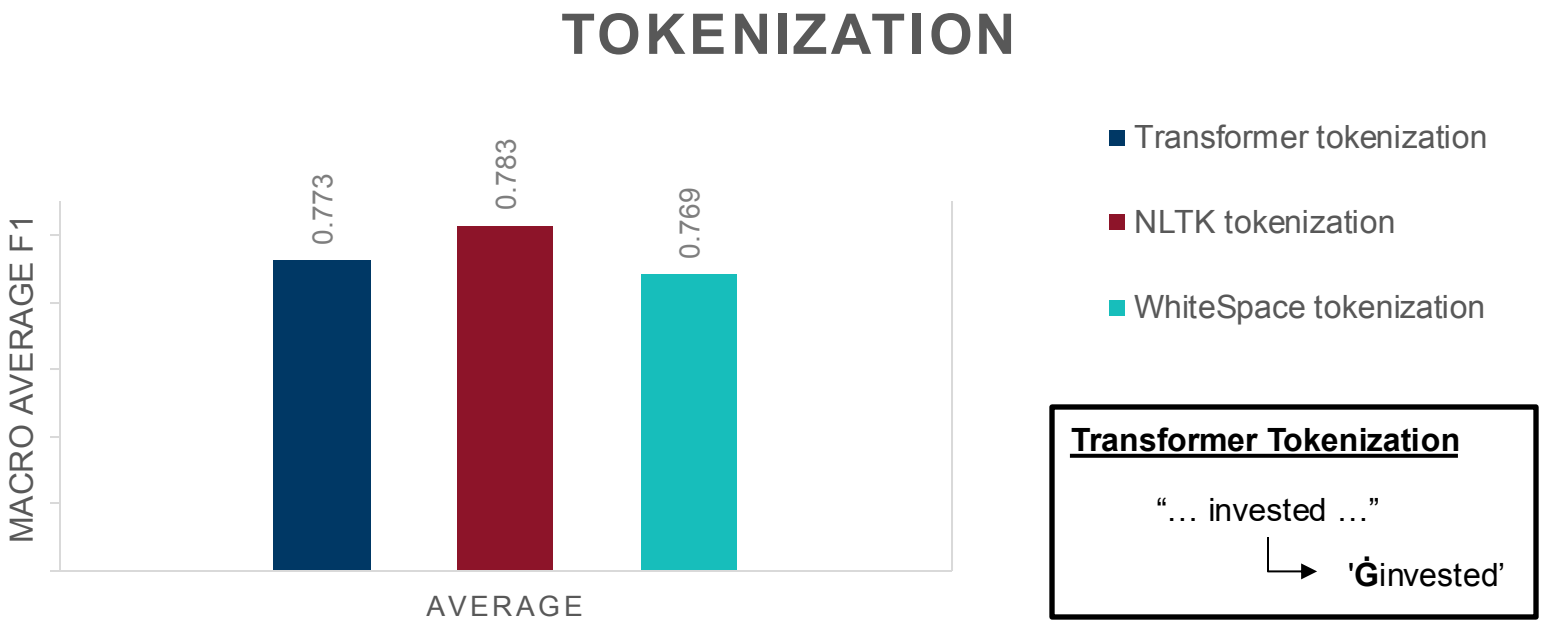


Tokenization

Word embedding based NLP system

03 04



Transformer Tokenization

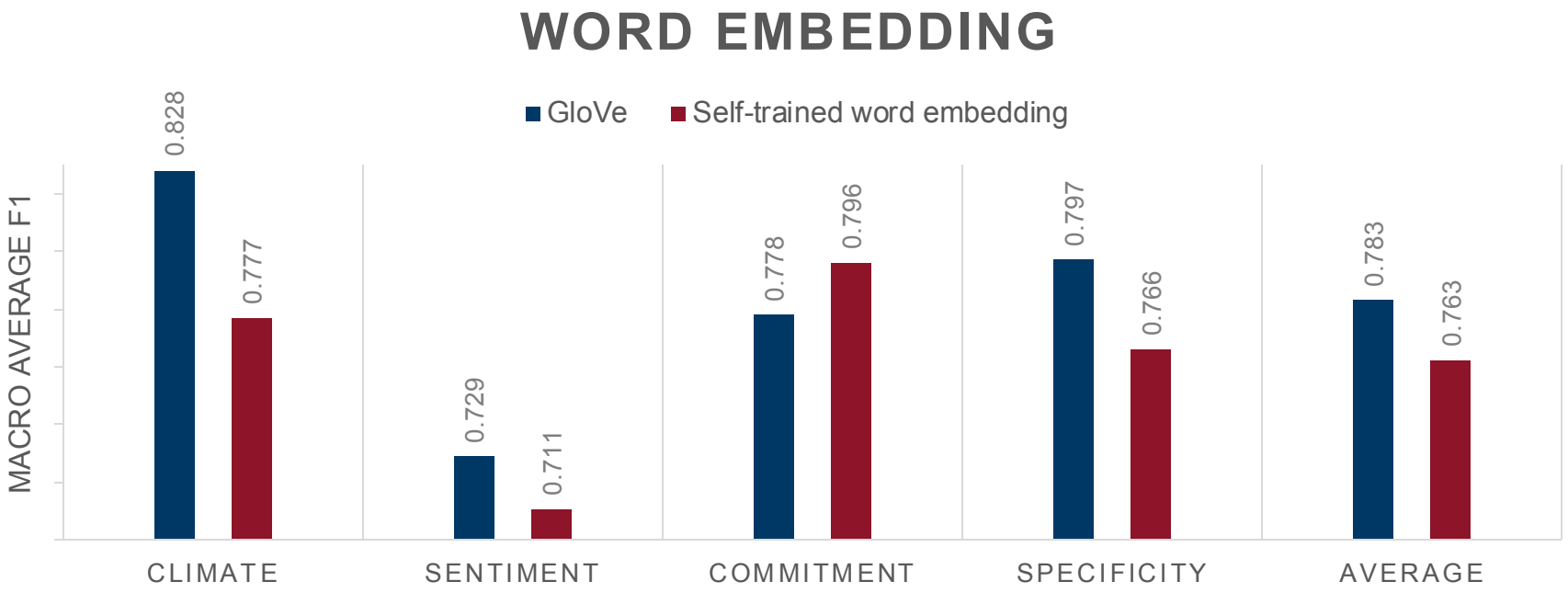
"… invested …"

→ 'Ġinvested'

Word embedding

Word embedding based NLP system

04 05



Model type (Cross-validated)

Word embedding based NLP system

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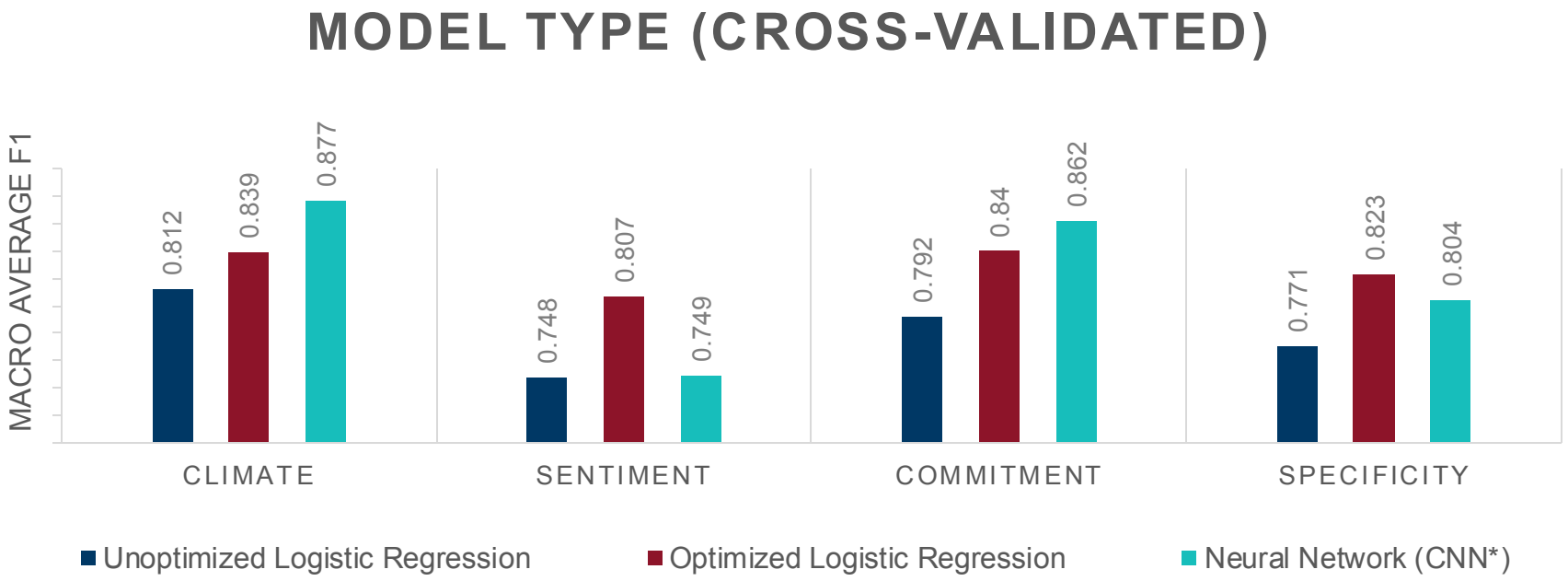
05 06



*CNN = Convolutional Neural Network

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Transfer Learning NLP System

Steps taken in building transfer learning based NLP system

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Transfer Learning NLP System

01 Finding pretrain models

- Apply filters on Hugging face – model
  - DistilRoBERTa-base model
  - ClimateBERT model

02 Data preprocessing

- AutoTokenizer.from_pretrained(Distil RoBERTa-base)
- Finetune token's max_length

03 Hyperparameter tuning

- Tuning batch size and number of epochs

04 Evaluation

On both tuned and not-tuned model

05 Cross validation

- On finetuned ClimateBERT model

Transfer learning by applying pre-trained models

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Usage of DistilRoBERTa-base model

01 02

Finding appropriate pretrain models

1. Apply these filters on Hugging face – model

- Task: Text classification
- Libraries: Transformer
- Languages: English

→ **DistilRoBERTa-base** is one of the most popular models (download and like) until November 2022

→ Other use cases have shown that this model is useful

Data preprocessing: finetune max_length of tokens:

```
tokens = tokenizer(batch['text'], truncation=True,
 max_length=512)
```

Max_length	Macro F1 for climate label
10	0.55
50	0.82
100	0.91
512	0.92

Train 4 models for 4 labels with DistilRoBERTa-base

Train data: project_training.json

Test data: project_validation.json

Set seed 0 to have reproducible result

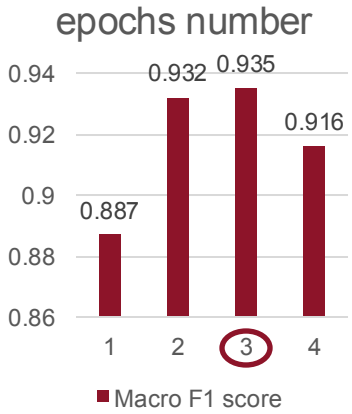
Transfer learning by applying pre-trained models

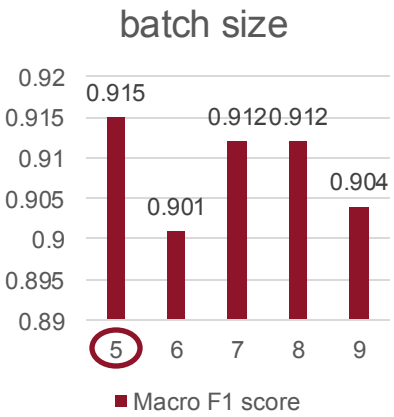
Usage of DistilRoBERTa-base model – Evaluation

03 04

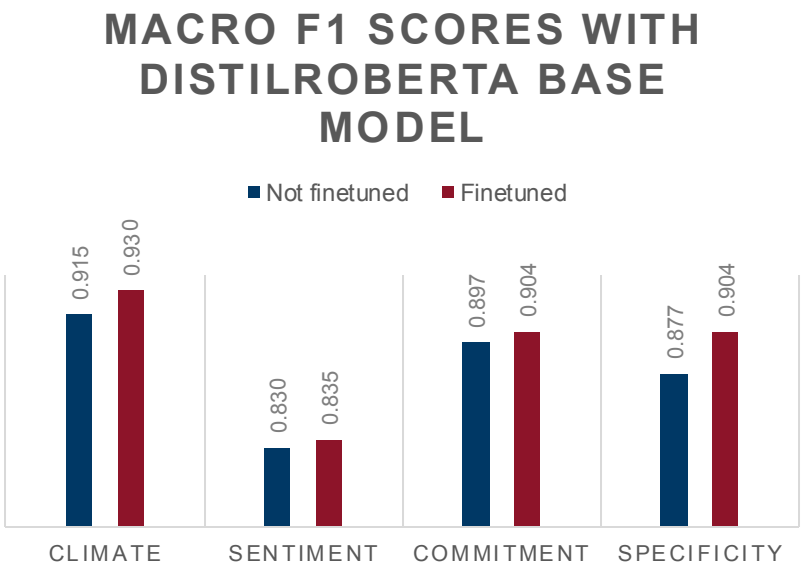
Hyperparameter tuning

- Default setting: num_epochs = 3, batch_size = 8, learning_rate = 5e-5, seed = 0
- Tuned the batch size and number of epochs





Evaluation results



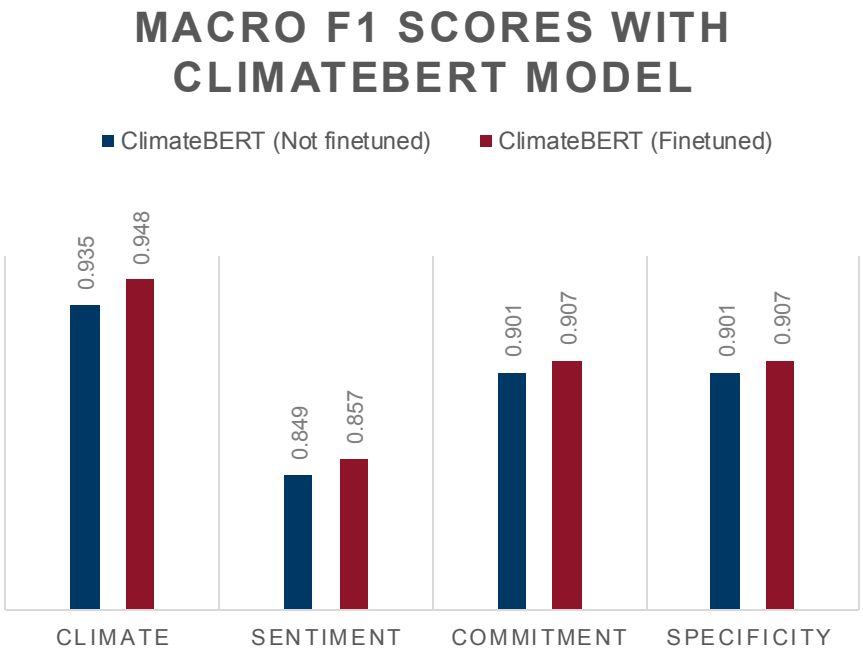
Transfer learning by applying pre-trained models

Usage of ClimateBERT model

ClimateBERT model

- Uses the **DistilRoBERTa model** as starting point
- Additionally **pre-trained on climate-related text**

  → *might included our test data in their training data, which would artificially inflate the performance of the model!*
- Macro F1 scores from all labels are higher with this model compared to the DistilRoBERTa-base model.*



* Not-tuned ClimateBERT has better Macro F1 scores in every label than not-tuned DistilRoBERTa-base; The same is for tuned ClimateBERT and tuned DistilRoBERTa-base.

Transfer learning by applying pre-trained models

Cross Validation

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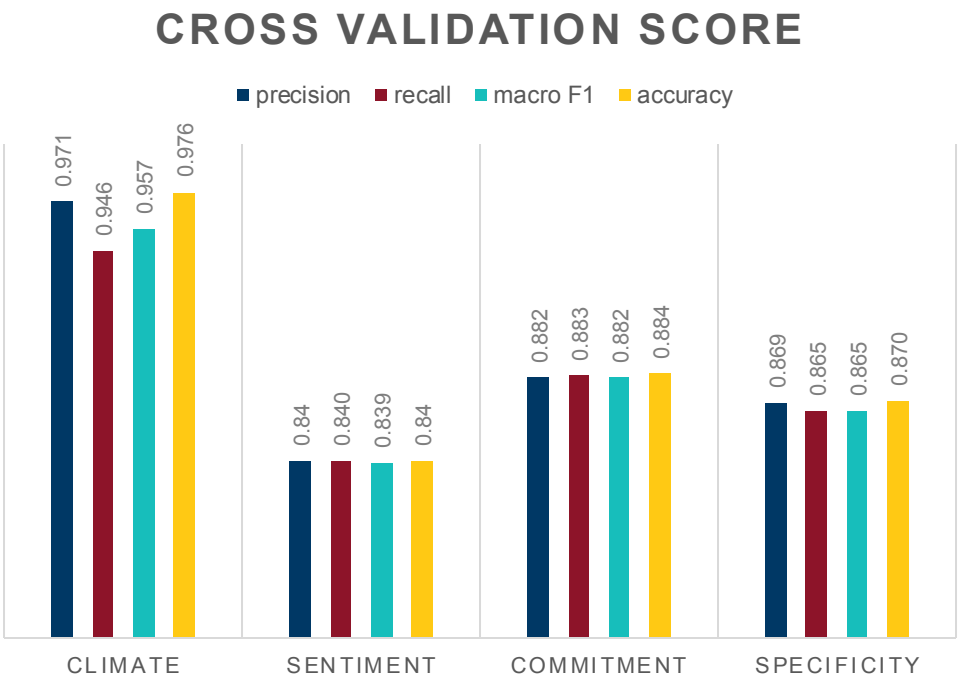
Conduction cross validation in fine-tuned ClimateBERT

- Climate macro F1 score increases
- The other three labels decrease slightly

→ We have robust models



Evaluation summery

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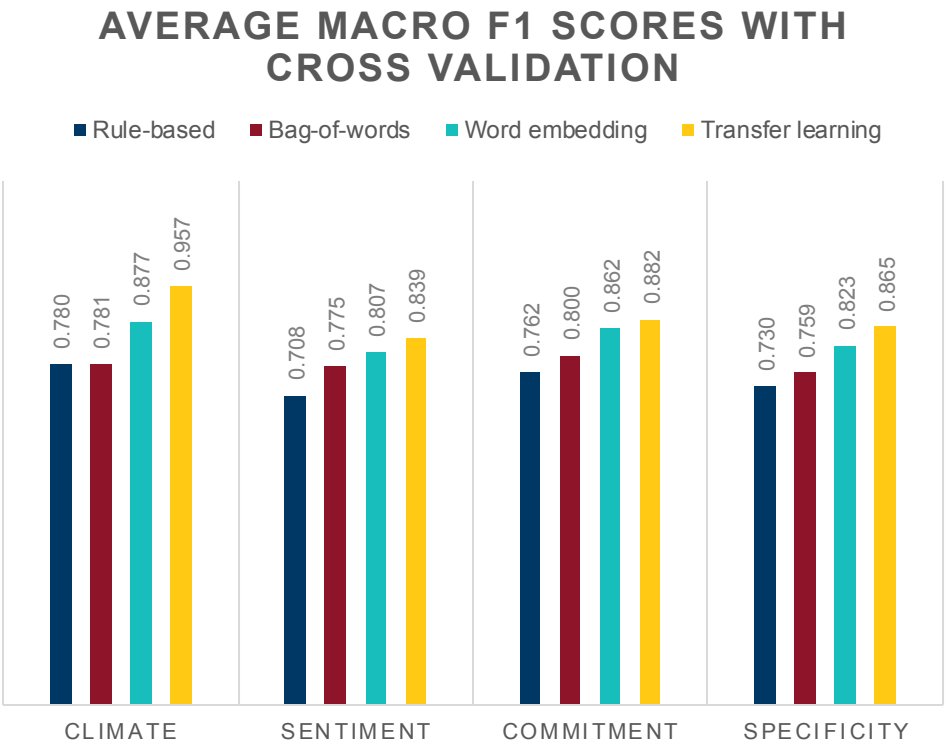
Transfer learning model has the highest macro F1 scores in all labels

04

1. The rule-based model has the lowest score for all labels

2. The transfer learning ClimateBERT model has the highest score for all labels



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Data Boosting

Finding similar text

Data Boosting

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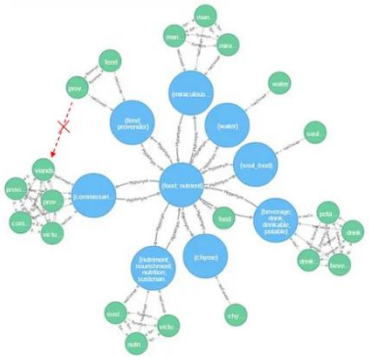
Step 1: Find 81 paragraphs that are misclassified wrongly in at least one label using ClimateBERT model

Step 2: We chose Knowledge-based Measures to find similar texts, which assess the semantic relatedness of words using a semantic network.



Sample of WordNet Graph

Step 3: find similar text in the 1 mio. unlabeled text

```
for 1 misclassifed paragraph in 81 misclassifed paragraphs:
    A = 1 misclassifed paragraph
    randomly pick 300 paragraphs in the 1 mio. unlabeled text

    for 1 unlabeled paragraph in 300 paragraphs:
        B = 1 unlabeled paragraph
        calculate the similarity scores between A and B
        save this score in a matrix

Find the top 3 similar paragraphs for each misclassifed paragraph
```

Step 4: manually label these similar texts

- Ask two people to label the same paragraphs
- Exclude 2 unsuitable paragraphs
- Results in 241 labelled paragraphs
- They are not included in initiate train and test data
- Balance data (climate label ratio: 4.9 → 1.5)

Train and Evaluation

Data Boosting

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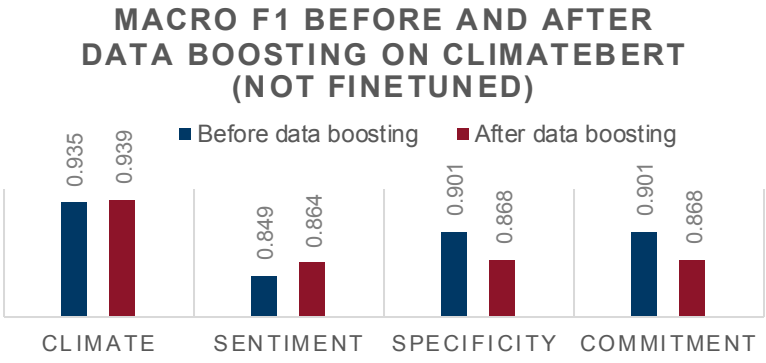
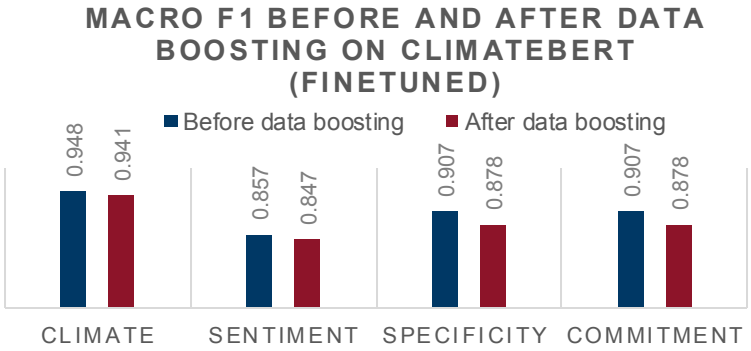
Train

Add the 241 extra paragraphs to the initial 400 training text, and test on the 400 test text

Evaluation

- Not tuned model: climate and sentiment are higher
- Tuned model: all labels have worst performances





Achievements | Outlook

Conclusion

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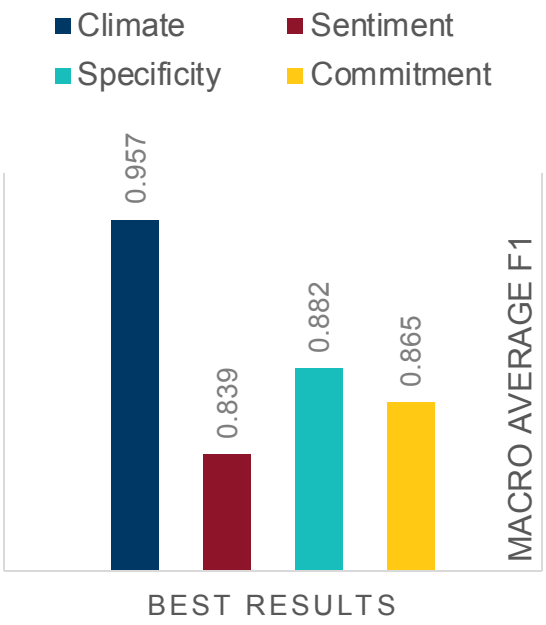
Best Results



Highlights of our Project

- Organizational:
  Weekly team meetings
- Achieved Tasks
  - ✓ Rule-based Model
  - ✓ Bag-of-Words Model
  - ✓ Word Embedding Model (2 Models)
  - ✓ Transfer Learning (2 Models)
  - ✓ Comprehensive Evaluation Metrics
  - ✓ Cross-Validation
  - ✓ Data Boosting

Outlook

- Rule-based:
  - Include more rules
- Word Embedding:
  - Apply ELMo, …
- ClimateBERT
  - Reasons for the good result
- Include XAI

Feedback and Questions

Data boosting

Which texts are misclassifed?

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- Climate label

```
[20] misclassified
0s

(array([119, 146, 151, 159, 172, 208, 216, 236, 267, 271, 289, 314, 325,
        332]),)
```

```
# misclassified # old

(array([ 24, 151, 169, 188, 208, 216, 236, 267, 289, 314, 325, 332]),)
```

Transfer learning by applying pre-trained models

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Another approach

- **1 model for the climate label, 1 model for the other 3 labels**
- Reason: there might be some relationships between the 3 other labels

Step 1: Convert sentiment label

sentiment_oppo	sentiment_neu	sentiment_risk
0	1	0
0	1	0
0	1	0
1	0	0
0	0	1
...	...	...
0	0	1
0	1	0
0	1	0
0	0	1
0	1	0

Step 2: Aggregate labelling columns

	text	labels
0	The accelerator programs have sub-portfolios o...	(1, 1, 0, 1, 0)
2	Climate change Climate change exposes UPM to v...	(1, 1, 0, 0, 1)
3	Several tools and methodologies aimed at asses...	(0, 1, 0, 0, 1)
4	We worked with the UK government to accelerate...	(0, 0, 1, 0, 0)
5	(changes in consumption behavior, failure of i...	(1, 1, 0, 0, 1)
...	...	...
393	The main impacts of the 4°C scenario were as f...	(1, 1, 0, 0, 1)
394	Emission sources not reported This section of ...	(1, 0, 0, 1, 0)
396	We have also signed up to the Partnership for ...	(0, 1, 0, 0, 1)
397	Suzano also is involved and spearheads externa...	(0, 1, 0, 1, 0)
399	UBS is also involved in other activities to re...	(0, 0, 0, 1, 0)

332 rows × 2 columns

Transfer learning by applying pre-trained models

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Another approach

- **1 model for the climate label, 1 model for the other 3 labels**
- Reason: there might be some relationships between the 3 other labels

Step 3: Train with multilabel classification model

```
model = MultiLabelClassificationModel('distilbert', 'distilbert-base-uncased', num_labels=5, args={"reprocess_input_data": True,
        "overwrite_output_dir": True,
        "num_train_epochs": 5,
        'train_batch_size':2})
```

Step 4: Predict the result

```
{'LRAP': 0.9110604525498143, 'eval_loss': 0.36393039726785253}
```

LRAP: Label ranking average precision

Data Boosting

Train data ratio

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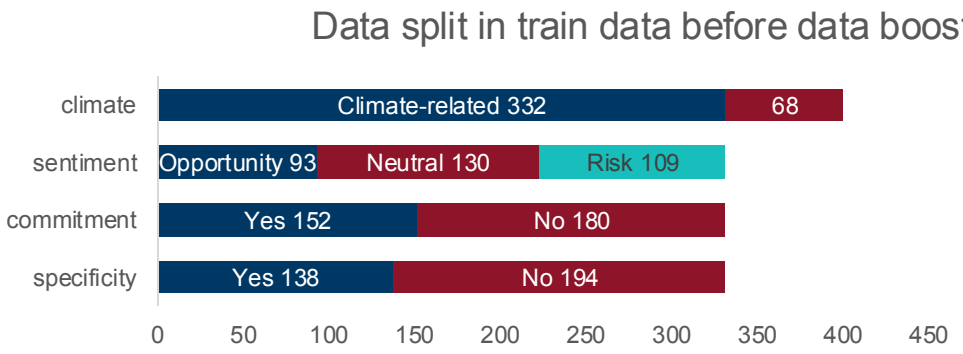

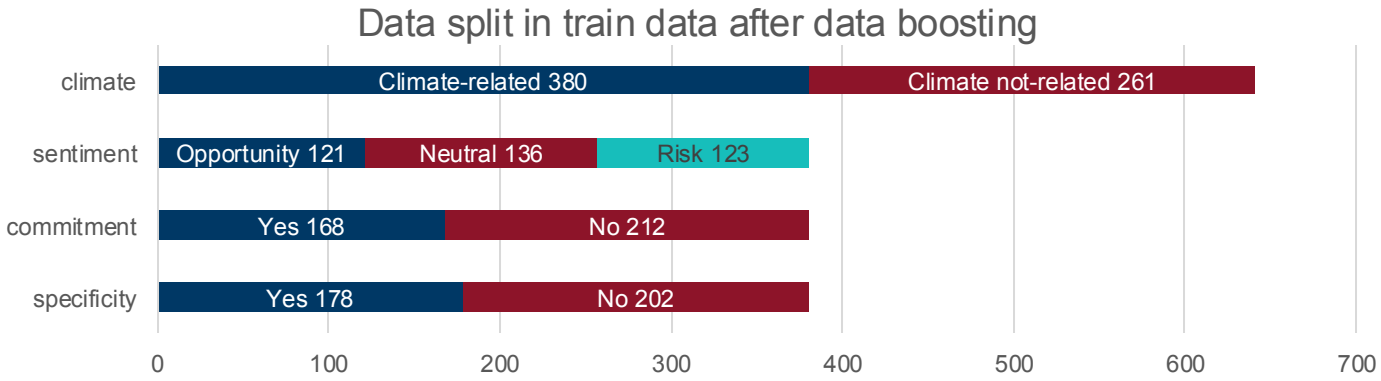

- `checkpoint = "distilroberta-base"`
- `tokenizer = AutoTokenizer.from_pretrained(checkpoint)`

Climate 5,4

```
0.9153439153439153
              precision    recall  f1-score   support

           0       0.96      0.98      0.97       329
           1       0.91      0.82      0.86        71

    accuracy                           0.95       400
   macro avg       0.93      0.90      0.92       400
weighted avg       0.95      0.95      0.95       400
```

```
5 4
0.9298961844967473
              precision    recall  f1-score   support

           0       0.98      0.96      0.97       329
           1       0.85      0.93      0.89        71

    accuracy                           0.96       400
   macro avg       0.92      0.95      0.93       400
weighted avg       0.96      0.96      0.96       400
```

Sentiment 8,4

```
0.8300573300573301
              precision    recall  f1-score   support

           0       0.76      0.81      0.79        81
           1       0.84      0.76      0.80       136
           2       0.88      0.94      0.91       112

    accuracy                           0.83       329
   macro avg       0.83      0.84      0.83       329
weighted avg       0.83      0.83      0.83       329
```

```
8 4
0.8354714219176259
              precision    recall  f1-score   support

           0       0.85      0.74      0.79        81
           1       0.79      0.84      0.81       136
           2       0.89      0.91      0.90       112

    accuracy                           0.84       329
   macro avg       0.84      0.83      0.84       329
weighted avg       0.84      0.84      0.84       329
```

- Commitment 9,3

```
0.8969190600522193
              precision    recall  f1-score   support

           0       0.89      0.87      0.88       139
           1       0.91      0.92      0.91       190

    accuracy                           0.90       329
   macro avg       0.90      0.90      0.90       329
weighted avg       0.90      0.90      0.90       329
```

```
0.9042140556176452
              precision    recall  f1-score   support

           0       0.86      0.92      0.89       139
           1       0.94      0.89      0.92       190

    accuracy                           0.91       329
   macro avg       0.90      0.91      0.90       329
weighted avg       0.91      0.91      0.91       329
```

- Specifity 9,3

```
0.8766866566716642
              precision    recall  f1-score   support

           0       0.83      0.90      0.86       139
           1       0.92      0.86      0.89       190

    accuracy                           0.88       329
   macro avg       0.87      0.88      0.88       329
weighted avg       0.88      0.88      0.88       329
```

```
0.9042140556176452
              precision    recall  f1-score   support

           0       0.86      0.92      0.89       139
           1       0.94      0.89      0.92       190

    accuracy                           0.91       329
   macro avg       0.90      0.91      0.90       329
weighted avg       0.91      0.91      0.91       329
```

- `checkpoint = "climatebert/distilroberta-base-climate-f"`
- `tokenizer = AutoTokenizer.from_pretrained(checkpoint)`

Climate 5,3

0.9354276305169018

	precision	recall	f1-score	support
0	0.98	0.98	0.98	329
1	0.90	0.89	0.89	71
accuracy			0.96	400
macro avg	0.94	0.93	0.94	400
weighted avg	0.96	0.96	0.96	400

0.948051948051948

	precision	recall	f1-score	support
0	0.98	0.98	0.98	329
1	0.93	0.90	0.91	71
accuracy			0.97	400
macro avg	0.95	0.94	0.95	400
weighted avg	0.97	0.97	0.97	400

P 0.9709947415737856
r 0.945693383786263
f1 0.9573442214572987
accuracy 0.97625

Sentiment 5,3

0.8493707874512828

	precision	recall	f1-score	support
0	0.77	0.85	0.81	81
1	0.85	0.81	0.83	136
2	0.93	0.90	0.91	112
accuracy			0.85	329
macro avg	0.85	0.85	0.85	329
weighted avg	0.85	0.85	0.85	329

5 3
0.8573848459397592

	precision	recall	f1-score	support
0	0.76	0.86	0.81	81
1	0.88	0.80	0.84	136
2	0.92	0.93	0.92	112
accuracy			0.86	329
macro avg	0.85	0.86	0.86	329
weighted avg	0.86	0.86	0.86	329

0.8401832618630506
0.8402904246312068
0.8387455656035392
0.8396331738437002

- Commitment 8,4

```
0.900709219858156
              precision    recall  f1-score   support

           0       0.87      0.90      0.89       139
           1       0.92      0.91      0.91       190

    accuracy                           0.90       329
   macro avg       0.90      0.90      0.90       329
weighted avg       0.90      0.90      0.90       329
```

```
0.9072298669072862
              precision    recall  f1-score   support

           0       0.87      0.92      0.90       139
           1       0.94      0.90      0.92       190

    accuracy                           0.91       329
   macro avg       0.91      0.91      0.91       329
weighted avg       0.91      0.91      0.91       329
```

```
0.8822366886192364
0.8832575248871176
0.8815530772660864
0.8835383914331283
```

- Specifity 8, 4

```
0.900709219858156
              precision    recall  f1-score   support

           0       0.87      0.90      0.89       139
           1       0.92      0.91      0.91       190

    accuracy                           0.90       329
   macro avg       0.90      0.90      0.90       329
weighted avg       0.90      0.90      0.90       329
```

```
0.9072298669072862
              precision    recall  f1-score   support

           0       0.87      0.92      0.90       139
           1       0.94      0.90      0.92       190

    accuracy                           0.91       329
   macro avg       0.91      0.91      0.91       329
weighted avg       0.91      0.91      0.91       329
```

```
0.868535516632900
0.864705998145145
0.864729757726222
0.869879243563454
```